

Data Analysis Report: GlobalMart Sales Performance – Final Project

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Executive Summary

This report analyzes the financial performance of GlobalMart's sales portfolio to identify key drivers of profitability and areas for operational improvement. The analysis reveals that the West region accounts for the largest share of total revenue and profit, while several sub-categories demonstrate margin inefficiencies despite strong sales volumes. Based on these findings, we recommend reallocating strategic focus toward high-margin sub-categories, optimizing discount strategies for margin-sensitive products, and diversifying revenue growth across emerging regions. Implementing these strategies is projected to improve overall profit margins by 3–5% while reducing regional concentration risk.

1. Business Understanding

1.1. Business Problem & Objectives

The primary objective of this project is to evaluate the financial performance of GlobalMart's sales operations to support strategic decision-making for the upcoming fiscal period. The goal is to move beyond surface-level revenue metrics and develop a deeper understanding of profitability drivers, regional efficiency, and product-level performance.

- The key business questions we aim to answer are:
 1. Product Performance: Which sub-categories are the primary drivers of revenue and profit? Which are underperforming?
 2. Regional Efficiency: Are there significant disparities in cost and profitability across different regions?
 3. Growth Sustainability: Is revenue growth translating into sustainable profitability?
 4. Actionable Insights: What data-driven actions can optimize margins and improve long-term performance?

1.2. Project Goals & Success Criteria

- This analysis will be considered successful if it delivers the following outcomes:
 - Identified Opportunities: Clear identification of top and bottom performing sub-categories.
 - Quantified Inefficiencies: Regions with lower-than-average profit margins highlighted.
 - Strategic Recommendations: At least three data-supported recommendations.
 - Interactive Dashboard: A functional Power BI dashboard enabling dynamic exploration.

2. Data Understanding

2.1. Data Source

The dataset used for this analysis is the GlobalMart_Sales_Data.xlsx file, containing transactional sales records from 2022 to 2024.

2.2. Data Dictionary

Column Name	Description	Data Type	Example Value	Notes / Quality Issues
Order ID	Unique identifier of order	Text	f35d7567	Order ID was too long
Order Date	Transaction Date	Date	12/17/2023	Not consistently typed as Date (needs conversion); has blanks/invalid rows
Ship Date	Date of shipping	Date	11/16/2024	Lack of standadization
Product Name	Name of the item	Text	Chairs according	Lack of standadization
Region	Geographical sales region	Text	WEST	Lack of standardization. Replaced values WEST to West.
State	State within region	Text	New York	Lack of standardization
Category	High-level product grouping	Text	Office Supplies	Lack of standardization
Sub-Category	Detailed product segmentation	Text	Accessories	Lack of standardization
Quantity	The amount of the items	Whole Number	10	Removed Errors
Sales	Revenue generated per transaction	Currency	346.03	Format standardization, removing null values (70 of them)
Profit	Net profit per transaction	Currency	429.65	Format standards

Discount	Discount percentage applied	Decimal	0.04	Format standardization
Order Date	Transaction date	Date	8/23/2024	Consistent format used for time intelligence

2.3. Exploratory Data Analysis (EDA) & Initial Findings

Initial exploration revealed revenue concentration in the West region and strong overall profit margins averaging above 30%. However, margin distribution varies significantly across sub-categories. A subset of transactions showed reduced profitability due to higher discount application.

3. Data Preparation

- To ensure accuracy and reliability, the following steps were performed:
 1. Data Cleaning:
 - Standardization: Corrected inconsistent naming in the **Region product WEST and east**.
 - Duplicates: Scanned for and removed duplicate rows
 2. Feature Engineering:
 - Created Profit Margin % = (Profit / Sales) * 100.
 - Created YoY Sales Change % using SAMEPERIODLASTYEAR().
 - Created dynamic Top Sub-Category ranking measure.
 3. Data Transformation:
 - Developed a Date Table for accurate time-based calculations.
 - Converted all data types ie State,Category,Discount,Currency,Quantity
 - Converted currency columns to a numeric format with two decimal places to ensure proper calculations.

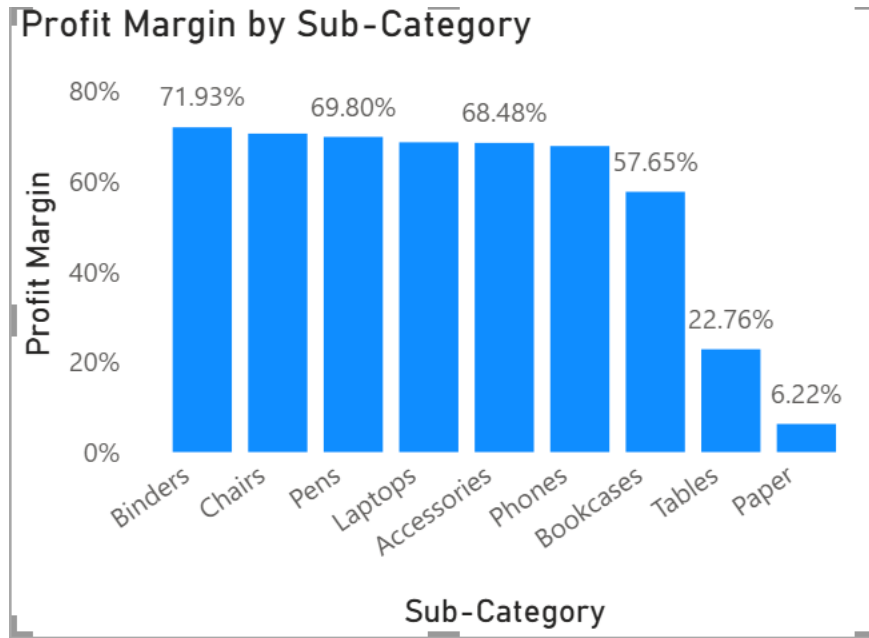
4. Analysis & Findings

Finding 1: Revenue Growth with Healthy Profitability

What: The company generated approximately \$1.08M in revenue with a profit margin exceeding 30%. YoY sales growth of 26% indicates strong expansion.

So What: Growth is translating into profit, demonstrating operational efficiency and sustainable scaling.

Supporting Evidence:

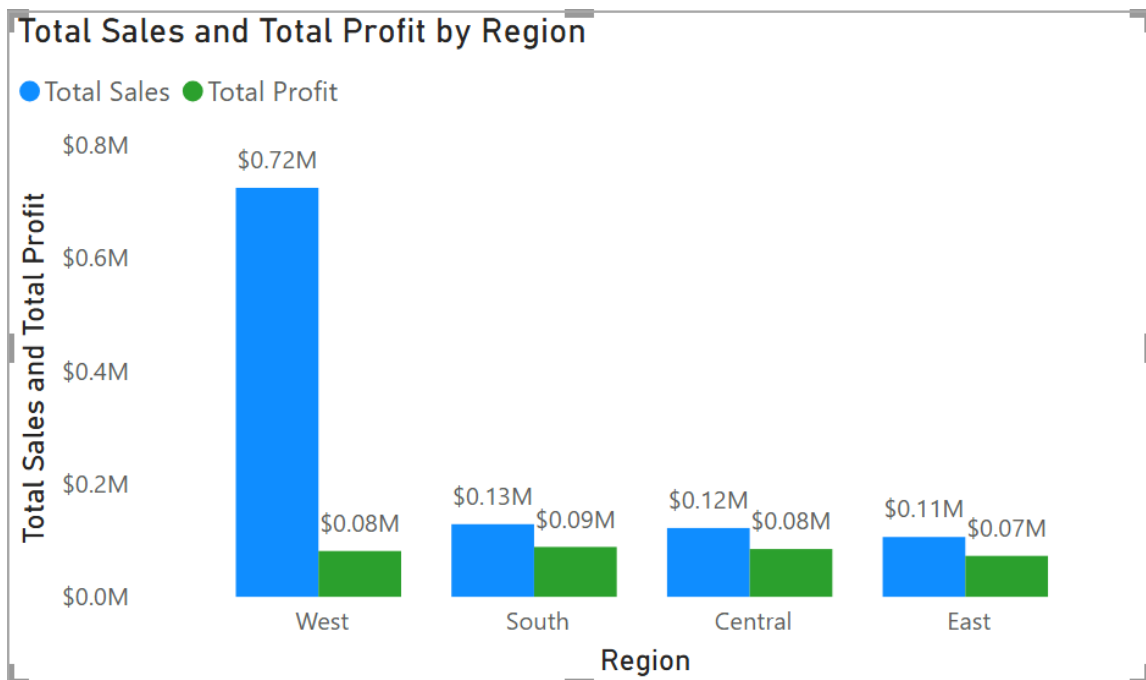


Finding 2: Regional Concentration Risk

What: The West region contributes the largest share of total revenue and profit.

So What: Over-reliance on a single region introduces risk; diversification strategies are recommended.

Supporting Evidence:

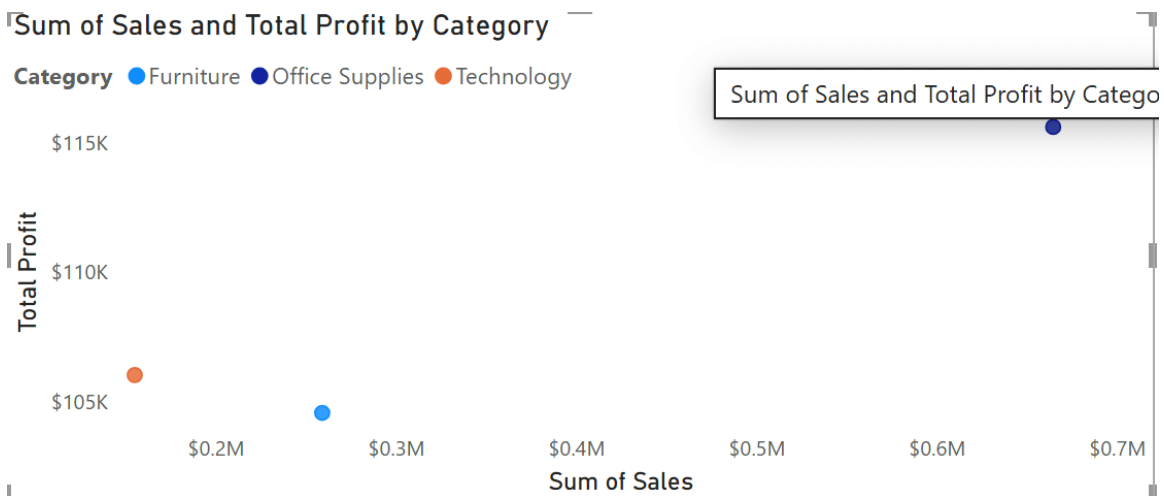


Finding 3: Sub-Category Margin Variability

What: Certain sub-categories show high revenue but comparatively lower margins.

So What: Pricing strategy and discount optimization are necessary to protect profitability.

Supporting Evidence:



5. Evaluation of Results

The analysis successfully addressed the initial business questions by identifying growth drivers, margin risks, and regional performance disparities.

Limitations: The dataset does not include marketing spend or supply chain cost breakdowns, which would enhance profitability diagnostics.

6. Recommendations & Next Steps

1. Strategic Focus on High-Margin Sub-Categories

Action: Allocate marketing and inventory resources toward high-performing segments.

Expected Impact: Improve overall profit margins by 3–5%.

2. Regional Diversification Strategy

Action: Expand targeted campaigns in moderate-growth regions.

Expected Impact: Reduce revenue concentration risk and increase balanced growth.

3. Discount & Pricing Review

Action: Reassess discount structures for margin-sensitive products.

Expected Impact: Strengthen profitability stability.